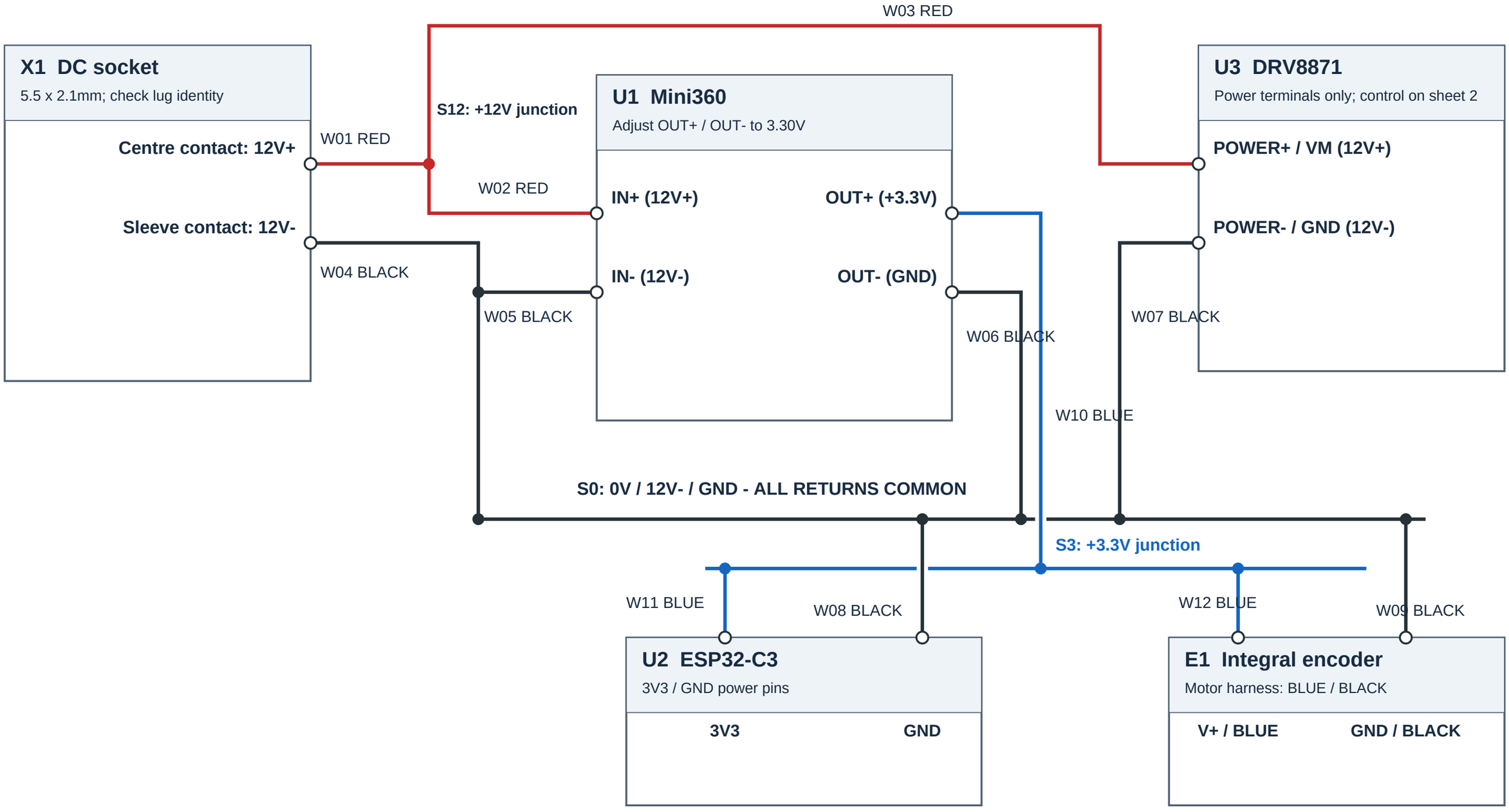
DETAILED WIRING - 12V / 3.3V POWER DISTRIBUTION

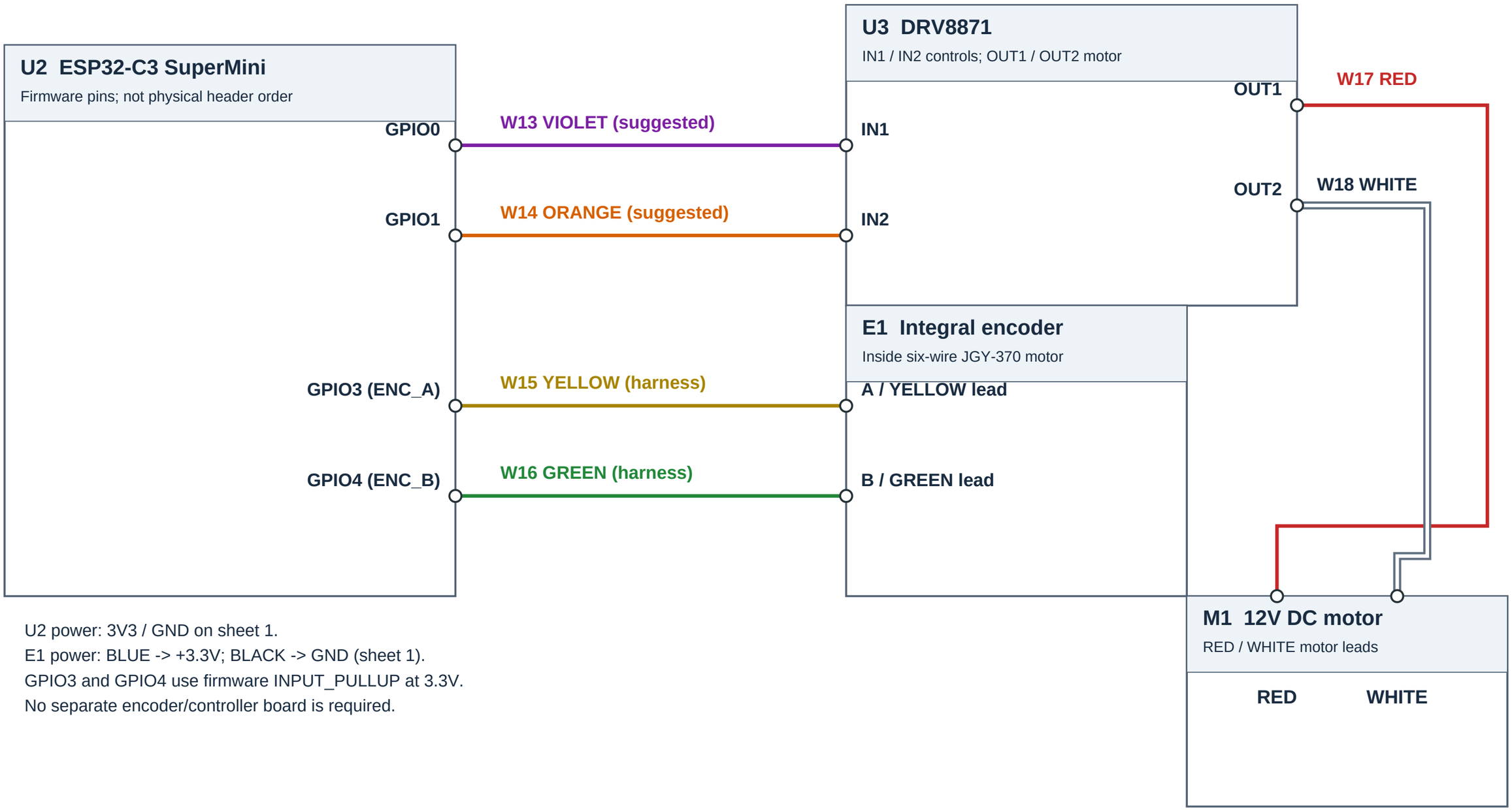
RB-EL-001 / REV B / 14 OCT 2026 / Firmware 2.0.2 / millimetres not applicable - schematic



Solid dots = connected junctions. Crossing lines without a dot are NOT connected.
S12 / S0 / S3 are drawing labels for electrical joins, not extra terminal blocks in the BOM.
RED / BLACK / BLUE hookup-wire colours are suggested; E1 BLUE and BLACK are the confirmed harness colours.
X1 switched third contact is unused; identify centre/sleeve with a meter, not by lug position.

DETAILED WIRING - MOTOR DRIVE / ENCODER FEEDBACK

RB-EL-001 / REV B / 14 OCT 2026 / Firmware 2.0.2 / millimetres not applicable - schematic

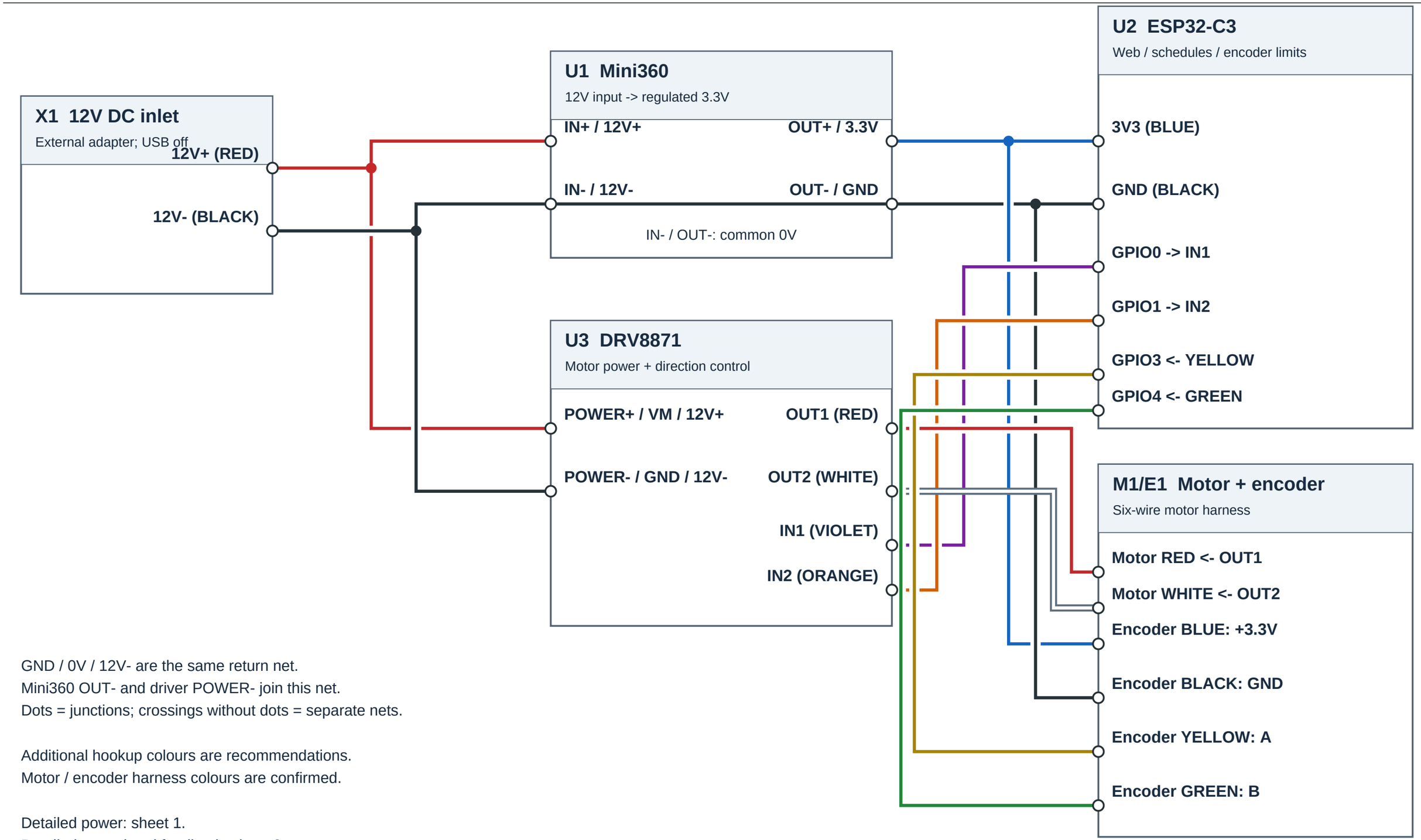


U2 power: 3V3 / GND on sheet 1.
E1 power: BLUE -> +3.3V; BLACK -> GND (sheet 1).
GPIO3 and GPIO4 use firmware INPUT_PULLUP at 3.3V.
No separate encoder/controller board is required.

OUT1 / OUT2 reverse polarity with direction: WHITE is NOT a permanent 12V- / GND lead.
Encoder BLACK = common 0V. Never apply +12V to the encoder supply or signal leads.

HIGH-LEVEL OVERVIEW - LABELLED TERMINALS AND COLOURS

RB-EL-001 / REV B / 14 OCT 2026 / Firmware 2.0.2 / millimetres not applicable - schematic



GND / 0V / 12V- are the same return net.
Mini360 OUT- and driver POWER- join this net.
Dots = junctions; crossings without dots = separate nets.

Additional hookup colours are recommendations.
Motor / encoder harness colours are confirmed.

Detailed power: sheet 1.
Detailed control and feedback: sheet 2.